

Computability & Logic, Exercises to Lectures 13+14

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Theoretical Exercises

From John C. Martin book

Manual

Make sure you do at least a few exercises of each block.

The suggested minimal material:

- 4.1, 4.3 (a),
- 4.30 (a), 4.38 (b)
- 4.49 (b), 4.50 (a)
- CYK parsing

Bonus material:

- 4.10(d), 4.3(c), (4.12 is difficult, maybe just the strategy?)
- 4.26 (b) (easy, regular grammars, but only if time)
- 4.54(a) (last step of Chomsky normal form)
- 4.14 (test your proof skills)
- 4.30(b) (test your puzzle/counting skills)

All Exercises

Context-Free Grammars

- 4.1 (a,e)
- 4.3 (a,c)
- 4.10 (b,d)
- 4.12 (not so easy, but a nice puzzle! The trick is to write this language as a union/concatenation of simpler languages, and combine their grammars)
- 4.14. Prove a stronger statement with induction on the length of the derivation.

Derivation tree and Ambiguity

- 4.30
- 4.38 (b)

Chomsky Normal Form:

- 4.49 (b)
- 4.50 (a)
- 4.54 (a)

CYK parsing. Take the Chomsky-Normal Form of 4.54a, and show the CYK-matrix for parsing the word $((()))$. Also show the derivation tree that is implicit in this CYK-matrix.

Regular Grammar

- 4.26 (b)